RYDEX

Ride Booking & Courier Management Platform

Project Progress Presentation

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Team Project • Full-Stack Web Application

1. Project Overview

One platform for transportation and shipment requirements

- RYDEX combines multi-vehicle ride booking with courier management in a single platform.
- Users can book different vehicle categories such as bikes, cars, autos, vans, trucks and lorries.
- The platform supports pickup/drop selection, ride requests, OTP verification, payments and live tracking.
- Users can also send couriers, estimate shipment cost and track parcels using a unique tracking ID.
- Partners and admins manage operational workflows, verification, checkpoints and platform monitoring.

2. Problem Statement & Objectives

Problem Statement

Conventional ride-booking applications mainly focus on passenger transportation. But users may also need different vehicle types to move goods, furniture or other items and may not have direct contacts for suitable transport.

RYDEX addresses this by bringing multiple vehicle categories and courier services into one platform.

Objectives

- Provide multiple vehicle categories
- Simplify ride booking and tracking
- Support secure OTP verification
- Enable cash and online payment
- Provide courier cost estimation
- Generate courier tracking IDs
- Support checkpoint-based parcel tracking
- Provide partner/admin management

3. Main Users & Responsibilities

USER

Book rides
Select vehicle type
Choose pickup/drop
Track active rides
Make payment
Create/send courier requests
Track courier using tracking ID

PARTNER

Receive ride requests
Accept / reject rides
Start and complete rides
Share live location
Verify pickup/drop OTP
Monitor earnings
Add courier checkpoints where authorized

ADMIN

Manage platform
Onboard and verify partners
Review partner information
Create/manage couriers
Add checkpoints
Monitor rides and courier operations
View analytics and messages

4. Multi-Vehicle Ride Booking

A single booking platform for different transportation needs

Bike

Select the appropriate vehicle category based on the user's transportation requirement.

Car

Select the appropriate vehicle category based on the user's transportation requirement.

Auto

Select the appropriate vehicle category based on the user's transportation requirement.

Van

Select the appropriate vehicle category based on the user's transportation requirement.

Truck

Select the appropriate vehicle category based on the user's transportation requirement.

Lorry

Select the appropriate vehicle category based on the user's transportation requirement.

5. User Ride Booking Flow

1. Pickup

Select pickup on map

2. Drop

Select destination

3. Vehicle

Choose suitable vehicle

4. Book

Confirm ride/payment

5. Partner

Request accepted

6. Track

Live location + status

7. Verify

Pickup/drop OTP

8. Complete

Ride completed

6. Partner Workflow & Earnings

- Partners receive ride requests and can accept or reject them.
- After acceptance, the partner can navigate to pickup and start the ride.
- Live location is shared during the active ride so the user can track the vehicle.
- Pickup and drop OTPs provide verification points in the ride lifecycle.
- Partners can monitor their earnings from completed rides.
- Current business rule: 90% of applicable ride revenue goes to the partner and 10% to the admin.

7. Partner Onboarding & Verification

Admin-controlled activation workflow

Stage 1

Basic Information

Stage 2

Documents

Stage 3

License

Stage 4

Screening

Stage 5

Verification

Stage 6

Video KYC

Stage 7

Admin Review

Stage 8

Active Partner

Admin reviews submitted details and pricing information before activating the partner.

8. Real-Time Tracking & Verification

Live Location

Partner location is updated during the active ride, allowing the user and partner to see the vehicle position on the map.

Ride Status

The booking progresses through request, acceptance, pickup, ride-start and completion states.

OTP Verification

Pickup and drop OTPs provide verification checkpoints and help ensure that the ride is started and completed correctly.

9. Courier Management

A second service available through the Send Courier option

Courier Creation

Courier information includes:

- Shipment type
- Product type
- Weight
- National / international shipment
- Delivery type — standard, same-day or overnight

The system uses internal pricing rules to calculate the estimated courier cost.

Courier Lifecycle

1. Courier details are collected
2. Cost is estimated
3. Admin creates the courier
4. Unique tracking ID is generated
5. Checkpoints are added during delivery
6. User tracks current status and location

10. Courier Tracking & Checkpoints

- Each created courier receives a unique tracking ID.
- Users and admins can use the tracking ID to identify and monitor the shipment.
- Authorized admins or partners can add checkpoints during delivery.
- Each checkpoint represents the parcel's current location or delivery state.
- Users can view checkpoint updates and follow the courier's progress until delivery completion.

Example Tracking Flow

Courier Created
↓
Tracking ID Generated
↓
Checkpoint 1
↓
Checkpoint 2
↓
Out for Delivery
↓
Delivered

11. Admin Dashboard & Monitoring

- Monitor ride bookings and overall platform activity.
- Manage partner onboarding and verification.
- Review partner pricing and submitted information.
- Manage courier creation and delivery checkpoints.
- Monitor ride and courier operational performance.
- Manage Contact Us messages and other platform-level activities.

Analytics Focus

Ride bookings
Weekly performance
Courier delivery progress
Completed / pending operations
Partner activity
Platform-level monitoring

12. Technology Stack

Frontend

Next.js / React
Tailwind CSS
Redux

Backend

Node.js
Express.js
REST APIs

Database

MongoDB
Mongoose

Real-Time

Socket.IO
Live location updates

Payments / Media

Razorpay
Cloudinary

Maps / Services

Maps & geolocation APIs
Supporting services

13. Current Project Progress

- ✓ Multi-role user, partner and admin modules
- ✓ Multi-vehicle ride booking concept and workflow
- ✓ Pickup/drop location selection
- ✓ Ride request accept/reject flow
- ✓ OTP-based pickup and drop verification
- ✓ Live ride location and status tracking
- ✓ Cash and online payment workflow
- ✓ Partner onboarding and verification workflow
- ✓ Partner earnings and 90/10 revenue rule
- ✓ Courier management and tracking workflow defined
- → Remaining: end-to-end testing, UI refinement, deployment checks and final documentation

THANK YOU

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